

Networks Lab Assignment 3

Building TCP Sockets

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1. For client to server communication -

Source IP address = 127.0.0.1 Port = 44180
Destination IP address = 127.0.0.1 Port = 20000

```
-----  
Source Address: 127.0.0.1  
Destination Address: 127.0.0.1  
[Stream index: 0]  
Transmission Control Protocol, Src Port:  
Source Port: 44180  
Destination Port: 20000  
-----
```

For server to client communication -

Source IP address = 127.0.0.1 Port = 20000
Destination IP address = 127.0.0.1 Port = 44180

```
-----  
Source Address: 127.0.0.1  
Destination Address: 127.0.0.1  
[Stream index: 0]  
Transmission Control Protocol, Src Port:  
Source Port: 20000  
Destination Port: 44180  
-----
```

2. The connection is initialized by client sending SYN packet to the server

```
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1  
Transmission Control Protocol, Src Port: 44180, Dst Port: 20000, Seq: 0, Len: 0  
Source Port: 44180  
Destination Port: 20000  
[Stream index: 0]  
[Conversation completeness: Complete, WITH_DATA (31)]  
[TCP Segment Len: 0]  
Sequence Number: 0 (relative sequence number)  
Sequence Number (raw): 576077006  
[Next Sequence Number: 1 (relative sequence number)]  
Acknowledgment Number: 0  
Acknowledgment number (raw): 0  
1010 .... = Header Length: 40 bytes (10)  
Flags: 0x002 (SYN)  
Window: 65495  
[Calculated window size: 65495]  
Checksum: 0xfe30 [unverified]  
[Checksum Status: Unverified]  
Urgent Pointer: 0  
Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-Operation (NOP), Window scale  
[Timestamps]
```

The server then responds with an **ACK, SYN** packet. That is to acknowledge client connection and wait for acknowledgement from client.

```
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 20000, Dst Port: 44180, Seq: 0, Ack: 1, Len: 0
  Source Port: 20000
  Destination Port: 44180
  [Stream index: 0]
  [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 0]
  Sequence Number: 0 (relative sequence number)
  Sequence Number (raw): 137341230
  [Next Sequence Number: 1 (relative sequence number)]
  Acknowledgment Number: 1 (relative ack number)
  Acknowledgment number (raw): 576077007
  1010 .... = Header Length: 40 bytes (10)
  Flags: 0x012 (SYN, ACK)
  Window: 65483
  [Calculated window size: 65483]
  Checksum: 0xfe30 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-Operation (NOP), Window scale
  [Timestamps]
  [SACK/ACK analysis]
```

The client then responds with an **ACK** packet to the server, completing the handshake.

```
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 44180, Dst Port: 20000, Seq: 1, Ack: 1, Len: 0
  Source Port: 44180
  Destination Port: 20000
  [Stream index: 0]
  [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 0]
  Sequence Number: 1 (relative sequence number)
  Sequence Number (raw): 576077007
  [Next Sequence Number: 1 (relative sequence number)]
  Acknowledgment Number: 1 (relative ack number)
  Acknowledgment number (raw): 137341231
  1000 .... = Header Length: 32 bytes (8)
  Flags: 0x010 (ACK)
  Window: 512
  [Calculated window size: 65536]
  [Window size scaling factor: 128]
  Checksum: 0xfe28 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
  [Timestamps]
```

3. The closure of connection is also done using 3 packets. Initially a **FIN** packet is sent from client to server.

```
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 44180, Dst Port: 20000, Seq: 1170, Ack: 1143, Len: 0
  Source Port: 44180
  Destination Port: 20000
  [Stream index: 0]
  [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 0]
  Sequence Number: 1170 (relative sequence number)
  Sequence Number (raw): 576078176
  [Next Sequence Number: 1171 (relative sequence number)]
  Acknowledgment Number: 1143 (relative ack number)
  Acknowledgment number (raw): 137342373
  1000 .... = Header Length: 32 bytes (8)
  Flags: 0x011 (FIN, ACK)
  Window: 512
  [Calculated window size: 65536]
  [Window size scaling factor: 128]
  Checksum: 0xfe28 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
  [Timestamps]
```

The server then responds with acknowledgement and another **FIN** packet. (**FIN**, **ACK** packet)

```
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 20000, Dst Port: 44180, Seq: 1143, Ack: 1171, Len: 0
  Source Port: 20000
  Destination Port: 44180
  [Stream index: 0]
  [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 0]
  Sequence Number: 1143 (relative sequence number)
  Sequence Number (raw): 137342373
  [Next Sequence Number: 1144 (relative sequence number)]
  Acknowledgment Number: 1171 (relative ack number)
  Acknowledgment number (raw): 576078177
  1000 .... = Header Length: 32 bytes (8)
  Flags: 0x011 (FIN, ACK)
  Window: 512
  [Calculated window size: 65536]
  [Window size scaling factor: 128]
  Checksum: 0xfe28 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
  [Timestamps]
```

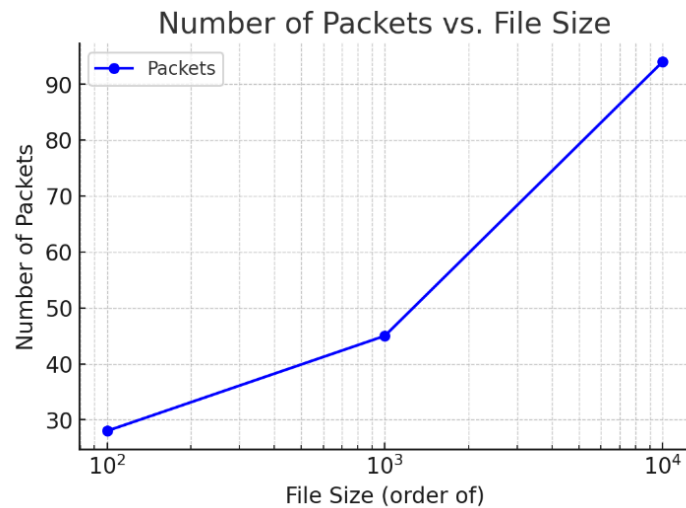
The client then finally responds with **ACK** packet marking end of socket connection.

```
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 44180, Dst Port: 20000, Seq: 1171, Ack: 1144, Len: 0
  Source Port: 44180
  Destination Port: 20000
  [Stream index: 0]
  [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 0]
  Sequence Number: 1171 (relative sequence number)
  Sequence Number (raw): 576078177
  [Next Sequence Number: 1171 (relative sequence number)]
  Acknowledgment Number: 1144 (relative ack number)
  Acknowledgment number (raw): 137342374
  1000 .... = Header Length: 32 bytes (8)
  Flags: 0x010 (ACK)
  Window: 512
  [Calculated window size: 65536]
  [Window size scaling factor: 128]
  Checksum: 0xfe28 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
  [Timestamps]
```

4. Inspected traffic -

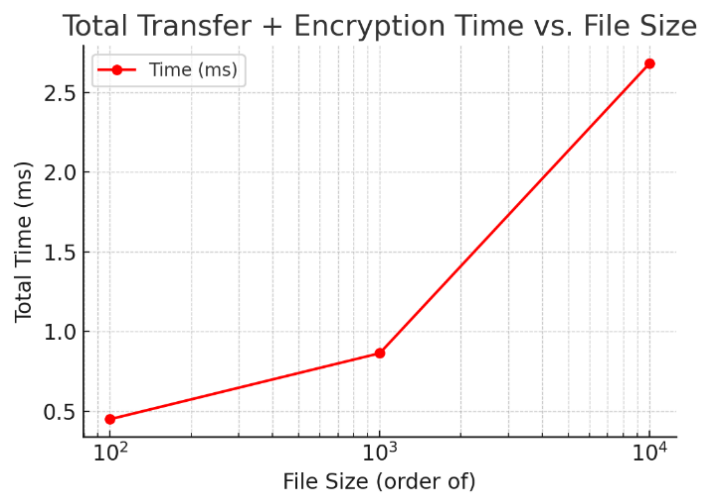
ip.src == 127.0.0.1 && ip.dst == 127.0.0.1						
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	127.0.0.1	127.0.0.1	TCP	74	44180 → 20000 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM TSval=1310739633 TSecr=0
2	0.000013	127.0.0.1	127.0.0.1	TCP	74	20000 → 44180 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM TSval=1310739633 TSecr=0
3	0.000024	127.0.0.1	127.0.0.1	TCP	66	44180 → 20000 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=1310699570 TSecr=1
8	0.002028	127.0.0.1	127.0.0.1	TCP	92	44180 → 20000 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=20 TSval=1310739633 TSecr=1
9	0.002051	127.0.0.1	127.0.0.1	TCP	66	20000 → 44180 [ACK] Seq=1 Ack=27 Win=65536 Len=0 TSval=1310739633 TSecr=
10	0.002720	127.0.0.1	127.0.0.1	TCP	166	44180 → 20000 [PSH, ACK] Seq=27 Ack=1 Win=65536 Len=100 TSval=1310739633 TSecr=
11	0.002729	127.0.0.1	127.0.0.1	TCP	66	20000 → 44180 [ACK] Seq=1 Ack=127 Win=65536 Len=0 TSval=1310739633 TSecr=
12	0.002741	127.0.0.1	127.0.0.1	TCP	166	44180 → 20000 [PSH, ACK] Seq=127 Ack=1 Win=65536 Len=100 TSval=1310739633 TSecr=
13	0.002763	127.0.0.1	127.0.0.1	TCP	66	20000 → 44180 [ACK] Seq=1 Ack=227 Win=65536 Len=0 TSval=1310739633 TSecr=
14	0.002778	127.0.0.1	127.0.0.1	TCP	966	44180 → 20000 [PSH, ACK] Seq=227 Ack=1 Win=65536 Len=900 TSval=1310739633 TSecr=
15	0.002784	127.0.0.1	127.0.0.1	TCP	66	20000 → 44180 [ACK] Seq=1 Ack=1127 Win=64640 Len=0 TSval=1310739633 TSecr=
16	0.002791	127.0.0.1	127.0.0.1	TCP	108	44180 → 20000 [PSH, ACK] Seq=1127 Ack=1 Win=65536 Len=42 TSval=1310739633 TSecr=
17	0.003195	127.0.0.1	127.0.0.1	TCP	66	20000 → 44180 [ACK] Seq=1 Ack=1169 Win=65536 Len=0 TSval=1310739633 TSecr=
18	0.003199	127.0.0.1	127.0.0.1	TCP	166	20000 → 44180 [PSH, ACK] Seq=1 Ack=1169 Win=65536 Len=100 TSval=1310739633 TSecr=
19	0.003223	127.0.0.1	127.0.0.1	TCP	66	44180 → 20000 [ACK] Seq=1169 Ack=101 Win=65536 Len=0 TSval=1310739633 TSecr=
20	0.003246	127.0.0.1	127.0.0.1	TCP	166	20000 → 44180 [PSH, ACK] Seq=101 Ack=1169 Win=65536 Len=100 TSval=1310739633 TSecr=
21	0.003254	127.0.0.1	127.0.0.1	TCP	66	44180 → 20000 [ACK] Seq=1169 Ack=201 Win=65536 Len=0 TSval=1310739633 TSecr=
22	0.003270	127.0.0.1	127.0.0.1	TCP	166	20000 → 44180 [PSH, ACK] Seq=201 Ack=1169 Win=65536 Len=100 TSval=1310739633 TSecr=
23	0.003293	127.0.0.1	127.0.0.1	TCP	66	44180 → 20000 [ACK] Seq=1169 Ack=301 Win=65536 Len=0 TSval=1310739633 TSecr=
24	0.003314	127.0.0.1	127.0.0.1	TCP	166	20000 → 44180 [PSH, ACK] Seq=301 Ack=1169 Win=65536 Len=100 TSval=1310739633 TSecr=
25	0.003322	127.0.0.1	127.0.0.1	TCP	66	44180 → 20000 [ACK] Seq=1169 Ack=401 Win=65536 Len=0 TSval=1310739633 TSecr=
26	0.003336	127.0.0.1	127.0.0.1	TCP	166	20000 → 44180 [PSH, ACK] Seq=401 Ack=1169 Win=65536 Len=100 TSval=1310739633 TSecr=
27	0.003343	127.0.0.1	127.0.0.1	TCP	66	44180 → 20000 [ACK] Seq=1169 Ack=501 Win=65536 Len=0 TSval=1310739633 TSecr=
28	0.003355	127.0.0.1	127.0.0.1	TCP	166	20000 → 44180 [PSH, ACK] Seq=501 Ack=1169 Win=65536 Len=100 TSval=1310739633 TSecr=
29	0.003362	127.0.0.1	127.0.0.1	TCP	66	44180 → 20000 [ACK] Seq=1169 Ack=601 Win=65536 Len=0 TSval=1310739633 TSecr=
30	0.003374	127.0.0.1	127.0.0.1	TCP	166	20000 → 44180 [PSH, ACK] Seq=601 Ack=1169 Win=65536 Len=100 TSval=1310739633 TSecr=
31	0.003380	127.0.0.1	127.0.0.1	TCP	66	44180 → 20000 [ACK] Seq=1169 Ack=701 Win=65536 Len=0 TSval=1310739633 TSecr=
32	0.003394	127.0.0.1	127.0.0.1	TCP	166	20000 → 44180 [PSH, ACK] Seq=701 Ack=1169 Win=65536 Len=100 TSval=1310739633 TSecr=
33	0.003400	127.0.0.1	127.0.0.1	TCP	66	44180 → 20000 [ACK] Seq=1169 Ack=801 Win=65536 Len=0 TSval=1310739633 TSecr=
34	0.003413	127.0.0.1	127.0.0.1	TCP	166	20000 → 44180 [PSH, ACK] Seq=801 Ack=1169 Win=65536 Len=100 TSval=1310739633 TSecr=
35	0.003419	127.0.0.1	127.0.0.1	TCP	66	44180 → 20000 [ACK] Seq=1169 Ack=901 Win=65536 Len=0 TSval=1310739633 TSecr=
36	0.003432	127.0.0.1	127.0.0.1	TCP	166	20000 → 44180 [PSH, ACK] Seq=901 Ack=1169 Win=65536 Len=100 TSval=1310739633 TSecr=
37	0.003440	127.0.0.1	127.0.0.1	TCP	66	44180 → 20000 [ACK] Seq=1169 Ack=1001 Win=65536 Len=0 TSval=1310739633 TSecr=

File Size (order of)	Number of Packets	Average Packet Size
100	28	98
1000	45	126
10000	94	285



5. Time for operations vs file size

File Size (order of)	Total time taken for transfer + encryption (ms)
100	0.449
1000	0.863
10000	2.68



6. Average packet size across all file sizes for data transmission is - 136 bytes